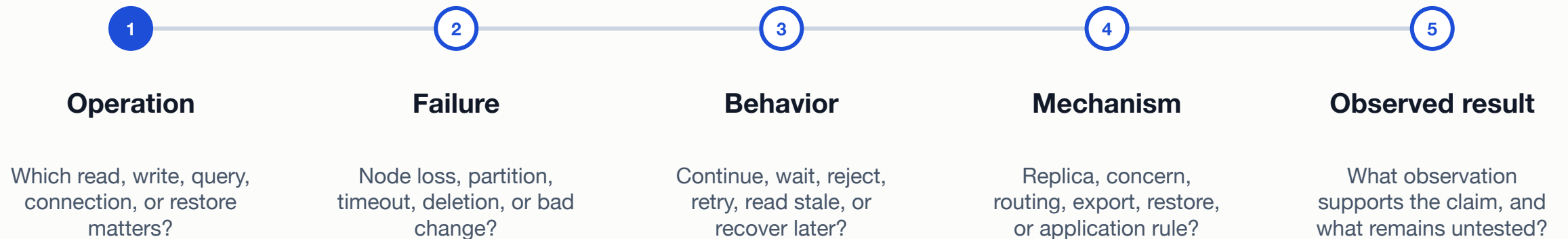


MongoDB Reliability and Recovery

Turn replication, read and write settings, and restore work into explicit operating promises.

A reliability promise needs five visible parts

Name the operation and failure before choosing the mechanism.



Replication and backup protect against different failures

Replication

- Maintains active copies from an ordered change history
- Supports continuity after some member failures
- Also reproduces valid but destructive writes and deletes

Backup or logical recovery artifact

- Preserves recoverable history outside active state
- Supports restoration after deletion, bad change, or loss
- Still requires a target, procedure, credentials, and verification

A replica set copies one ordered stream of changes



Members converge on the replica-set history; replication is asynchronous and failover takes time.

Write concern defines the acknowledgment a client waits for

A timeout can leave the application uncertain about whether the write exists.

REQUEST

The client asks for a required level of acknowledgment, such as majority.

COST

More acknowledgment can add latency or make success unavailable during failures.

AMBIGUITY

A timeout means the condition was not confirmed in time, not necessarily that nothing wrote.

Read preference selects a member; concern limits visibility

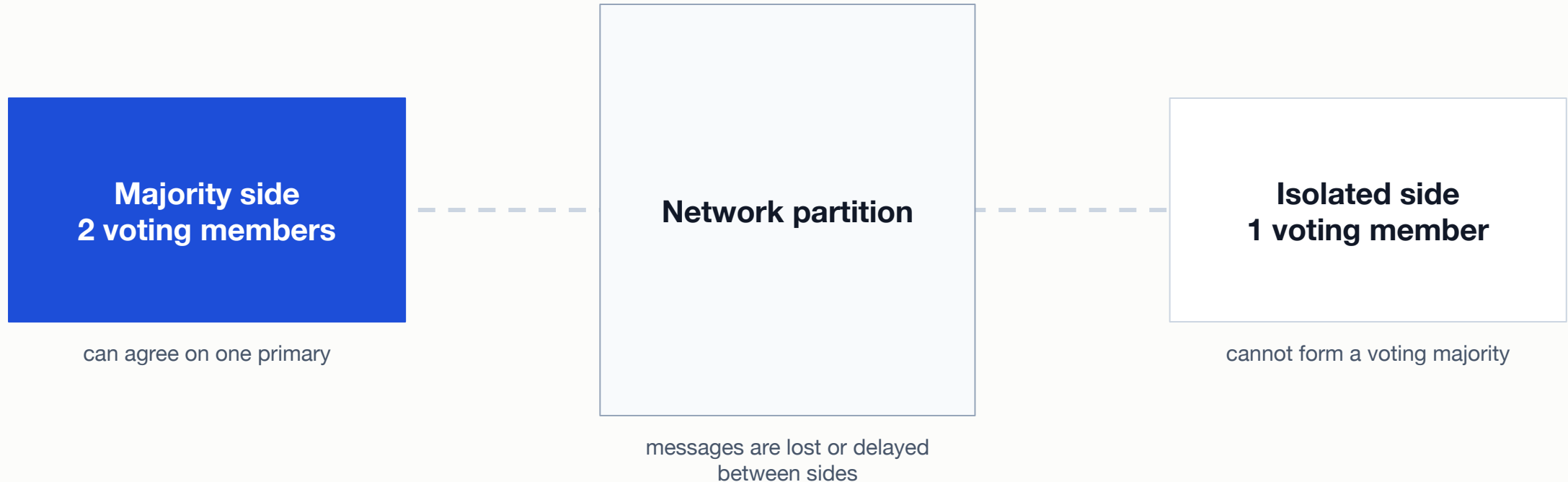
Read preference

- Selects eligible members such as primary or secondary-preferred
- Can serve locality or read-distribution goals
- A secondary may be behind the primary

Read concern

- Defines consistency and isolation properties of returned data
- Interacts with sessions, topology, and write concern
- Does not choose which member receives the read

A partition forces an operation-specific choice



Preserving one current write history can require rejecting or delaying operations on one side.

Reliability settings should follow user expectations

Workload expectation	Candidate decision	Mechanism	Cost or limit
Confirmed ticket survives minority member loss	majority acknowledgment	replica-set acknowledgment	latency and timeout ambiguity
Resident immediately sees submitted ticket	primary-oriented confirmation read	routing plus suitable visibility	primary load and reconnect delay
Hourly dashboard may be slightly stale	secondary-preferred where justified	read routing	replication lag
Accidental collection delete is recoverable	separate logical artifact	export, restore, verification	recovery point and recovery time

Lab 1: Match expectations to reliability decisions

Build one decision matrix, analyze one partition, and connect one workload to the final-project plan.

- For each Metro Support expectation, name the setting or mechanism, failure behavior, results, and limitation.
- Analyze the two-member majority side and one-member isolated side without assigning a permanent CAP label.
- Add one short workload checkpoint that links to, but does not redefine, the canonical final project.

SUBMIT ONE THING / one week_12_reliability_decision.md file

Recovery is a procedure, not a file

Export elsewhere, restore to a separate target, verify behavior, and state what the artifact omits.

Extended JSON preserves BSON meaning that JSON cannot

The representation must protect types before it can protect values.

ORDINARY JSON

Has strings, numbers, booleans, null, arrays, and objects.

BSON

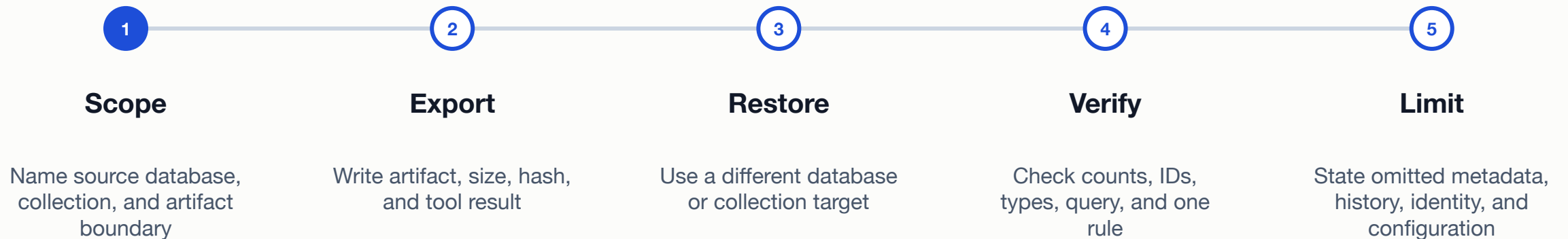
Adds database types such as ObjectId, date, decimal, binary, and integer variants.

EXTENDED JSON

Encodes BSON type meaning in a JSON-compatible representation.

A recovery drill closes the loop from source to restored state

Never overwrite the source while learning whether the artifact works.



JSON recovery and mongodump cover different scopes

Canonical Extended JSON exercise

- Readable collection-level interchange artifact
- Can preserve represented BSON value types
- Does not automatically recreate indexes, validators, users, or project settings

mongodump and mongorestore

- Database Tools logical dump and restore workflow
- Includes collection data and supported metadata such as options and indexes
- Still does not recreate every Atlas project, network, or identity control

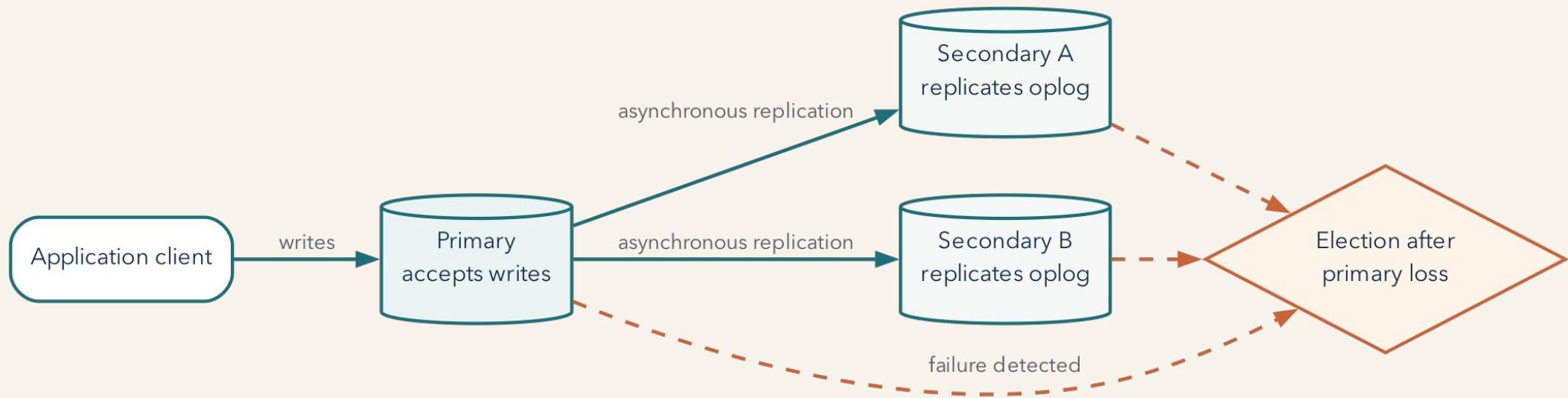
Lab 2: Restore MongoDB data and verify the claim

Run the recovery notebook through an Atlas connection or the complete offline path.

- Export canonical Extended JSON, record size and SHA-256, and restore into a different target.
- Verify counts, identifiers, BSON types, one meaningful query, and one expected rule failure.
- Compare the artifact with mongodump and confirm that no URI or private account detail appears in output.

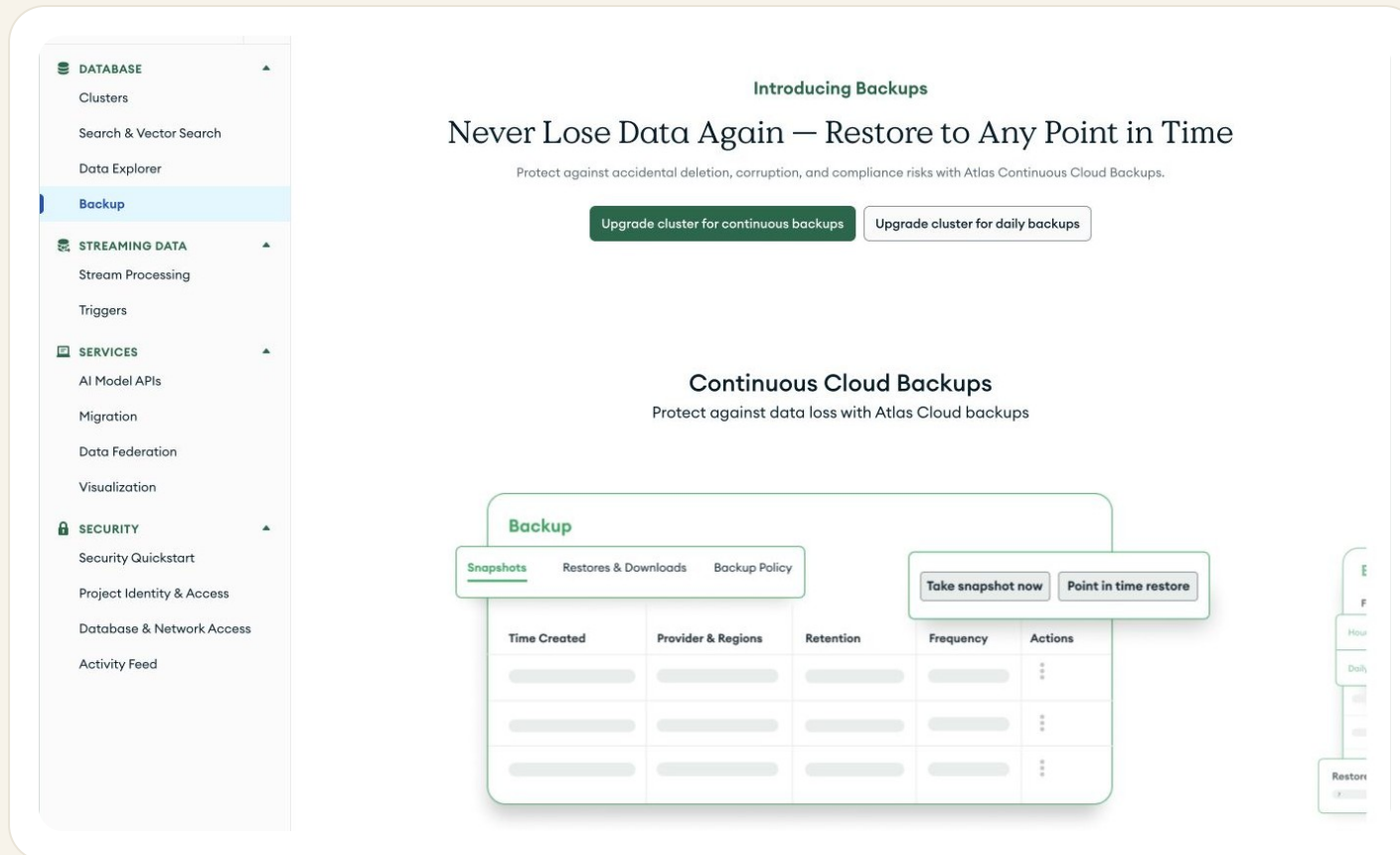
SUBMIT ONE THING / one completed 05_mongodb_logical_recovery.ipynb notebook

Replication maintains service state across members



replication supports continuity; it also copies authorized deletion

Atlas Free does not include managed continuous backups



1

A free classroom cluster needs a separate logical export path

2

Plan features can change; verify the current Atlas documentation

3

A copied file still requires a restore target and checks

Reliable systems make failure behavior and recovery procedures explicit

Replicas preserve service history; separate artifacts preserve recoverable history.

- Which operation and failure does the promise address?
- Which setting changes acknowledgment, routing, or visibility?
- Which restore checks confirm scope, and which omissions remain?